

The Derangement Proportion of a Transitive Group Is at Least $1/|\Omega|$: A Disproof of Conjecture 00000002311

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Abstract

The source asserts that the proportion of fixed-point-free elements of a transitive group on Ω is at least $c/|\Omega|$ with $c = \frac{1}{2}$, and that this constant is tight, as verified by Frobenius groups. The inequality is true, but the tightness clause is false. For every finite transitive group of degree $n \geq 2$ the proportion is at least $1/n$ (Cameron and Cohen). Hence no transitive group, Frobenius or not, attains or approaches $\frac{1}{2} \cdot \frac{1}{n}$, and the tight constant is 1; it is attained by the Frobenius group S_3 on three points. The general bound, the count of derangements in every Frobenius group, the example and the negation of the source's assertion are verified in Lean, using Burnside's lemma from Mathlib.

Reading of the source. Let a finite group G act transitively on a finite set Ω with $n = |\Omega|$. A *derangement* is an element of G without fixed points; D is the set of derangements and $\delta(G) = |D|/|G|$ is the derangement proportion. Both language versions of the source make two assertions:

- (i) $\delta(G) \geq c/|\Omega|$ with $c = \frac{1}{2}$;
- (ii) this constant is tight, and Frobenius groups verify the tightness.

The source names only one constant, $c = \frac{1}{2}$, and we read “the corrected constant” in (ii) as this constant. A constant in a lower bound is called tight when it cannot be replaced by a larger one. We consider three precise forms of (ii):

(T1) no constant $c' > \frac{1}{2}$ satisfies (i) for all transitive groups;

(T2) some Frobenius group has $\delta(G) = \frac{1}{2}/|\Omega|$;

(T3) for every $\varepsilon > 0$ some Frobenius group has $\delta(G) < (\frac{1}{2} + \varepsilon)/|\Omega|$.

We take $n \geq 2$ throughout. For $n = 1$ there are no derangements, so (i) itself fails; we do not use this degenerate case. A *Frobenius group* is a transitive permutation group of degree $n \geq 2$ in which no non-identity element fixes two points and some non-identity element fixes a point.

Theorem 1 (Cameron–Cohen). *Let a finite group G act transitively on a finite set Ω with $n = |\Omega| \geq 2$. Then $n|D| \geq |G|$, that is, $\delta(G) \geq 1/n$.*

Proof. Write $\text{fix}(g)$ for the number of fixed points of g . By Burnside's lemma, $\sum_{g \in G} \text{fix}(g)$ equals $|G|$ times the number of orbits on Ω , so

$$\sum_{g \in G} \text{fix}(g) = |G|. \tag{1}$$

Let G act on $\Omega \times \Omega$ by $g(x, y) = (gx, gy)$. A pair is fixed by g exactly when both entries are fixed, so g fixes $\text{fix}(g)^2$ pairs. The diagonal is invariant, and as $n \geq 2$ there is a pair off the diagonal; hence there are at least two orbits, and Burnside's lemma gives

$$\sum_{g \in G} \text{fix}(g)^2 \geq 2|G|. \tag{2}$$

For every g we have

$$(\text{fix}(g) - 1)(\text{fix}(g) - n) \leq n \cdot \mathbf{1}_D(g). \quad (3)$$

Indeed, if $g \in D$ the left side is $(-1)(-n) = n$. Otherwise $1 \leq \text{fix}(g) \leq n$, and the left side is at most 0. Summing (3) over G and using (1) and (2),

$$n|D| \geq \sum_g \text{fix}(g)^2 - (n+1) \sum_g \text{fix}(g) + n|G| \geq 2|G| - (n+1)|G| + n|G| = |G|. \quad \square$$

Theorem 2 (Frobenius groups). *Let G be a finite Frobenius group of degree n . Then $|D| = n-1$ and $\delta(G) = (n-1)/|G|$.*

Proof. The identity fixes n points and is not a derangement. A non-identity element fixes at most one point, so it fixes exactly one point unless it is a derangement. By (1), $|G| = n + (|G| - 1 - |D|)$, hence $|D| = n - 1$. $\square$

Corollary 1. *The following hold.*

1. *For every finite transitive group of degree $n \geq 2$, $\delta(G) \geq \frac{1}{n} > \frac{1}{2} \cdot \frac{1}{n}$. In particular (i) holds, also with the constant 1 in place of $\frac{1}{2}$.*
2. *(T1), (T2) and (T3) are false. So the conjunction of (i) and (ii) is false in each of the three readings of tightness.*
3. *The constant 1 is tight in all three senses: the Frobenius group S_3 acting on three points has $\delta = \frac{1}{3} = 1/n$.*

Proof. (1) is Theorem 1. For (2): the constant $c' = 1 > \frac{1}{2}$ satisfies (i), which refutes (T1). A Frobenius group is transitive of degree $n \geq 2$, so $\delta(G) \geq 1/n$; this is larger than $\frac{1}{2}/n$, which refutes (T2), and with $\varepsilon = \frac{1}{2}$ it refutes (T3). For (3): in S_3 a transposition fixes exactly one point and a 3-cycle fixes none, so S_3 is a Frobenius group with $|D| = 2$ and $\delta = 2/6 = 1/3$. Thus no $c' > 1$ satisfies (i), and equality holds for a Frobenius group. $\square$

Remark 1 (Not formalised). For a Frobenius group with point stabiliser of order h one has $|G| = nh$ and, by Theorem 2, $n\delta(G) = (n-1)/h$. Theorem 1 gives $h \leq n-1$; classically h divides $n-1$, because the stabiliser acts freely on the other $n-1$ points. So the value $n\delta(G)$ ranges over numbers $(n-1)/h \geq 1$ and equals 1 exactly when $h = n-1$, as for $\text{AGL}(1, q)$. Theorem 1 is due to P. J. Cameron and A. M. Cohen (Discrete Math. 106/107 (1992), 135–138), who also show that for permutation groups equality holds exactly for sharply 2-transitive groups. The Boston–Shalev conjecture, now a theorem of Fulman and Guralnick, concerns a lower bound for δ independent of n for simple groups; it is not addressed here.

The earlier submission and its error

Pull request #190 (orionsheep) addressed the same clause (ii). It computed that the Frobenius group $\text{AGL}(1, 7)$ of order 42 has exactly 6 derangements, so that $n\delta = 1$, and concluded that Frobenius groups do not verify tightness of $\frac{1}{2}$. It was closed without merging on 2026-10-03. The computation was correct. The reviewer named two errors, and both are real. First, the Lean file used Lean core only; it counted residues with a Boolean function on natural numbers, and its decisive theorem was

`theorem not_half_tight :  $\neg (6 = 3)$ .`

No group, action, derangement proportion or tightness statement occurred. Second, one Frobenius group with $n\delta = 1$ does not show that no other Frobenius group does better; this needs a general fact, which that submission mentioned only in prose and only for the groups $C_p \rtimes C_m$. The present submission proves the general fact for all transitive groups (Theorem 1) and for all Frobenius groups (Theorem 2), with Mathlib’s groups and actions, and states the tightness readings as propositions.

Formal verification and scope

In Lean, `derangements G Ω` is the set of g with $g \cdot x \neq x$ for all x , for a Mathlib `MulAction` (for the full symmetric group this is the defining formula of Mathlib’s own `derangements`), and `derangementProportion` is the quotient of the two cardinalities in $\mathbb{Q}$. Transitivity is Mathlib’s `MulAction.IsPretransitive`. Burnside’s lemma is Mathlib’s

`MulAction.sum_card_fixedBy_eq_card_orbits_mul_card_group,`

applied to Ω and to $\Omega \times \Omega$ with Mathlib’s product action. The theorem `cameron_cohen` is Theorem 1: for every group G and action on Ω with G and Ω finite, the action transitive and $2 \leq |\Omega|$,

`Nat.card G ≤ Nat.card Ω * Nat.card (derangements G Ω).`

The rational forms are

`one_div_card_le_proportion,      half_div_card_lt_proportion.`

The structure `IsFrobenius G Ω` has four fields: transitivity, $2 \leq |\Omega|$, that a non-identity element fixing x and y forces $x = y$, and the existence of a non-identity element with a fixed point. Theorem 2 is

`frobenius_card_derangements,      frobenius_proportion,`

and `frobenius_ne_half` states that no Frobenius group has $\delta = \frac{1}{2}/|\Omega|$. For S_3 , realised as `Equiv.Perm (Fin 3)` acting on `Fin 3`, the theorems `s3_isFrobenius` and `s3_proportion` prove the Frobenius property and $\delta = 1/3$.

The proposition `LowerBound c` is (i) with the constant c , quantified over all finite transitive actions of degree at least two. The three readings of tightness are

`CannotBeImproved c,    AttainedByFrobenius c,    ApproachedByFrobenius c.`

`Conjecture` is the conjunction of `LowerBound (1/2)` with the disjunction of the three readings at $c = \frac{1}{2}$. The final theorem `conjecture_false` is its negation; it is proved by refuting each of the three readings. The theorem `corrected_statement` proves the bound and all three readings of tightness for $c = 1$. Finite checks for S_3 use kernel evaluation by `decide`; no `native_decide`, `sorry` or custom axiom occurs.

Scope. This is a disproof of the source’s statement as a whole, through its tightness clause; clause (i) alone is true and is proved here in a stronger form. The actions in the Lean theorems are not assumed faithful, so they include all transitive permutation groups. The degree is assumed to be at least two. The Frobenius property is defined in the file by the permutation-group condition above, because Mathlib has no such notion; its equivalence with the kernel-and-complement description is not used. The remark is not formalised. If “the corrected constant” were meant to be a constant other than the stated $c = \frac{1}{2}$, the source would not say which; Corollary 1 shows that 1 is the only constant for which the bound and the tightness clause both hold. A reading of (ii) that does not concern the size of the constant is not addressed.

Source and reproduction

The exact bilingual source is included byte-for-byte as `SOURCE.md`. Its repository path is

`conjectures/00000002311.md`.

The project pins Lean 4.19.0 and Mathlib commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

From `lean/`, run `lake build`, then

`lake env lean -DwarningAsError=true Main.lean`.

The included public-Git manifest locks all dependency revisions. The `verification/` directory records the fresh-build logs, theorem-axiom reports, artifact hashes and the self-review. No supplementary numerical computation is required.